

Iron metabolism

Metabolismo do ferro

Ana Maria de Souza*

Abstract

The present paper deals with iron metabolism, from its absorption, regulatory factors, storage, and distribution to body compartments according to existing knowledge.

Keywords: Iron/metabolism; Ferritin; Transferrin

Resumo

O presente artigo trata do metabolismo de ferro desde sua absorção, fatores de regulação, estocagem e distribuição aos compartimentos corpóreos à luz do conhecimentos atuais.

Descritores: Ferro/metabolismo; Ferritina; Transferrina

Iron metabolism

Iron is an essential nutrient for survival of most living organisms and a component or co-factor of hundreds of proteins and enzymes. The biological importance of iron is widely attributed to its chemical properties. There are two oxidative forms, ferrous (Fe^{+2}) and ferric (Fe^{+3}) iron, and iron is able to accept and donate electrons, being useful in the catalytic center of fundamental biochemical reactions. In mammals, these reactions are performed by iron-containing proteins, iron-sulphur enzymes, hemoproteins and iron-containing enzymes (non-heme and non-iron-sulphur enzymes). The activity of most of these enzymes decreases in tissue iron deficiency⁽¹⁾.

Hemoproteins are involved in a variety of crucial biological processes, such as reversible binding of oxygen to hemoglobin and myoglobin, which are responsible for oxygen transport and storage, respectively; transport of electrons in oxidative phosphorylation process through cytochromes; oxygen metabolism through the enzymes oxydase, peroxydases, catalase and hydrolase.

Moreover, iron is a component of iron-sulphur and oxygenase non-heme proteins involved in molecular-oxygen controlled interactions and in a ribonucleotide reductase enzyme system required to convert ribose into deoxyribose of nucleic acid, and, consequently, to produce DNA and carry genetic information. Iron deficiency in cells delays growth and leads to cell death.

In a neutral pH, iron is practically insoluble and it is carried and processed by binding agents, such as transferrin and ferritin, or by low molecular weight compounds, such as citrate, aminoacids and adenosine 5'-triphosphate (ATP)⁽²⁾.

The living organisms should be able to control the internal amount of iron and to appropriately respond by changing its entry and storage processes. Failure in this control result in anemia or iron overload.

Throughout life, humans maintain a constant concentration of iron ranging from 35 to 45 mg/Kg. These levels are maintained by a positive iron balance and equilibrium between absorption and loss. The average intake of normal individuals is approximately 12-18 mg of iron/day and 1-2 mg are absorbed. The daily iron excretion in case of no bleeding is roughly one milligram, and it occurs mainly through the gastrointestinal tract by exfoliation of epithelial cells, bile secretion, and epidermal scaling, and also by the urinary tract. In the reproductive phase, women loose, in average, 2 mg of iron/day, due to menses⁽³⁾.

The absorption of iron is a regulated physiological process that serves as a primary mechanism to control iron balance. The most important stimuli for iron absorption are tissue iron storage, rate of erithropoiesis and tissue hypoxia. Absorption increases response to tissue hypoxia and production of erythrocytes. Apparently, this "erythropoietic regulator" involves a

* Associate Professor of the Department of Clinical, Toxicological and Bromatological Analyses, Faculdade de Ciências Farmacêuticas de Ribeirão Preto da Universidade de São Paulo - USP - Ribeirão Preto (SP), Brazil.

Corresponding author: Ana Maria de Souza. Av. do Café, s/nº, Ribeirão Preto, São Paulo (SP), Brazil.

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still unknown soluble sign or factor, which is released from the bone marrow and interferes in intestinal absorption, thus affecting the crypt cells programming⁽³⁻⁴⁾. The mechanism by which the duodenal crypt cells detect iron body storage is unknown. Recent studies suggest that this function is, at least, partly performed by hepcidin, which is a supposedly regulatory hormone that acts centrally in communicating body storage to absorptive cells. Hepcidin is a peptide highly expressed in hepatic cells and its expression is stimulated by iron overload⁽¹⁾.

One of the main iron hemostasis control points is in the duodenal epithelial cell layer. Iron entry regulation occurs in two enterocyte interfaces. The apical membrane is specialized in transporting heme- and ferrous iron to cells and the basolateral membrane in carrying iron to other parts of the body⁽⁵⁾.

Most iron in diet is in the ferric form. This form is soluble in acid pH and insoluble in alkaline pH, forming insoluble polyhydroxy groups that make iron biologically unavailable. In the stomach, iron is mobilized from food by chelation of mucins, ascorbate and histidine, among others. Heme protein iron is absorbed from food (meats) in a different and more efficient manner than inorganic iron. Heme-iron enters in the intestinal mucosa cells as an intact metalloporphyrin, after globin release by proteolytic enzymes. This entry is facilitated by heme receptors and a vesicular transport system. In the cell, Fe^{+2} is released from porphyrin by microsomal heme-oxygenase and enters into circulation like the inorganic form.

The intestinal cells have alternative forms to transport inorganic iron. There are separated pathways to transport ferrous and ferric forms that are known as IMP (integrin/mobilferrin/paraferritin) pathway for Fe^{+3} and DMT-1 (divalent metal transporter 1) pathway for Fe^{+2} ⁽³⁾.

Ferric iron enters the intestinal mucosa cell via a b3-integrin combined with mobilferrin. A large protein complex called paraferritin is formed containing b3-integrin, mobilferrin and flavin-monooxygenase. Paraferritin behaves as ferrireductase, reducing iron to ferrous iron that is available for iron-containing protein synthesis⁽³⁾.

DMT-1, also known as DCT-1 (divalent cation transporter) or Nramp-2 (natural resistance associated macrophage protein), mediates the active transport of iron, zinc, manganese, cobalt, cadmium, copper, nickel and lead. However, its main role is to transport iron by means of a proton-coupled process through the apical membrane in the cell⁽¹⁾.

Dietary Fe^{+3} must be reduced to Fe^{+2} by a duodenal ferric reductase - Dcytb (duodenal

cytochrome b) - before its transport via DMT-1. The expression of Dcytb, a transmembrane heme protein highly expressed in brush border membrane of enterocytes, is modulated by changes in physiological regulators of iron absorption⁽⁶⁾.

In the enterocyte, iron may be stored as ferritin or be transferred to plasma through the basolateral membrane. In normal conditions, iron is stored and kept in mucosal cells. In cases of iron deficiency or increased erythrocyte production, mucosal cells produce little ferritin and most iron entering the cell is available for transport through the basolateral membrane⁽¹⁾.

The Fe^{+2} transporter through the basolateral membrane was identified by three groups and called ferroportin or IREG-1 or MTP1. It is a transmembrane glycoprotein located in polarized basolateral membrane cells and expressed in the duodenum, placenta (iron transport from mother to embryo), liver Kupffer cells (iron recycling), spleen, macrophages and kidneys. The transporter requires an accessory protein bound to the membrane called hephaestin. It is a multicopper protein similar to ceruloplasmin, which acts as an intracellular ferroxidase. One of its major roles is to facilitate iron oxidation, enabling its fast binding with plasma transferring and delivery to cells that express transferring receptors, thus avoiding endothelial damage^(1,4).

Iron is distributed to body compartments according to their function. The major compartment comprises iron necessary for specific reactions. It is the operational compartment, which corresponds to 69% of the total body content and is primarily composed of hemoglobin and myoglobin, and of heme (cytochromes, peroxidases and catalase) and non-heme enzymes.

The second major compartment is storage, accounting for 30% of total body iron. Storage iron is found in hepatocytes and macrophages, especially in bone marrow, spleen and liver. This compartment comprises non-toxic forms of sequestered iron, such as ferritin and hemosiderin. It represents absorption greater than necessary for the operational compartment.

The storage and operational compartments are bound to a small transport compartment, which consists of transferring-bound iron, present in plasma or bound to cellular membrane receptors. Ferritin is a protein and its function is to sequester and store iron. Reversible formation and dissolution of a solid nanomineral-hydrated is the main function of ferritin. Moreover, ferritin detoxifies excess iron and, possibly oxygen and reactive oxygen⁽⁷⁾. It consists of an apoprotein capsule, with a molecular weight of 460KDa,

formed by 24 subunits of different types: L (*light*) and H (*heavy*) subunits. Iron is stored in the ferric form as ferric hydroxide phosphate, approximately composed of $(\text{FeOOH})_8 (\text{FeOPO}_3\text{H}_2)$. Ferritin may accommodate up to 4,500 iron atoms, thus doubling its molecular mass to 900 KDa⁽⁸⁾.

The H-rich isoferritins are found in cells with high iron demand for heme synthesis, such as cardiac muscle cells and erythroid precursors. The L-rich isoferritins predominate in storage-related tissues, such as liver and spleen. Modifications in the proportion of H and L subunits allow cells to adjust to iron metabolism. The increase in L subunit is associated with storage, whereas H subunit is more abundant when iron is needed for cell metabolism. Inflammatory and infectious conditions, response to xenobiotics and differentiation are some situations with altered subunit proportion⁽⁸⁾.

Cells with iron overload contain another type of storage – hemosiderin, which is probably a product of ferritin degradation. Apparently, in these conditions, ferritin is involved by liposomes, in which proteases promote a partial dissolution of the nucleus, resulting in formation of hemosiderin, an insoluble protein⁽⁸⁾.

Little ferritin circulates in plasma and its concentration is related to iron deposit levels. The origin and possible physiological role of plasma ferritin have not been explained yet; however, some studies in patients with L ferritin mutations suggest that serum ferritin is poor in iron, immunologically similar to L ferritin and may have a G subunit (glycosylated)⁽¹⁾.

Transferrin is a monomeric glycoprotein produced in the liver and consists of a polypeptide chain of 679 aminoacids in two homologue domains, N-terminal and C-terminal, each containing a Fe^{+3} binding site. Transferrin has three main functions: a) Fe^{+3} solubilization; b) high-affinity iron binding avoiding that the free metal produces free radicals and c) iron supply to cells, involving its interaction with membrane receptors, that is, transferring receptors^(1,9).

Except for mature erythrocytes, transferrin receptors (TfR) are expressed in all cells, and the higher concentrations are found in cells responsible for hemoglobin synthesis, placenta and fast-dividing cells. TfR could be detected in cell membranes by means of flow cytometry using the monoclonal antibody CD71⁽²⁾. Human TfR is a dimeric transmembrane glycoprotein composed of two identical subunits linked by disulfide. Each monomer consists of a phosphorylated cytoplasmic N-terminal domain, a hydrophobic transmembrane segment and an extracellular segment, including the C-terminal domain where three oligosaccharide chains are bound. Each unit is able to bind to one transferring molecule⁽²⁾.

In physiological pH, TfR affinity is higher for diferric transferrin, ensuring that TfR be preferentially saturated by iron-containing transferring that is required for cell metabolism. The complex is internalized by endocytosis mediated by receptors forming small lisosomal vesicles called endosomes.

In the first stage, transferrin (Tf) binds to specific receptors on cell surface through a physicochemical interaction and does not require an established temperature and energy expenditure. The Tf-TfR complex clusters in specific membrane areas known as coated pits. This region invaginates and losses its coating, forming a vesicle called endosome. Iron is released from transferrin in the interior of endosomes by a temperature- and energy-dependent process involving endosomal acidification. Subsequently, a non-identified ferrireductase reduces Fe^{+3} to Fe^{+2} and enables a divalent metal transporter (DMT-1) to transfer iron to cytoplasm through the endosomal membrane. Only the erythroid cells show some evidence of having a specific target for iron, which are the heme production sites by means of ferrochelatase in the mitochondria. At pH 5.5, apotransferrin remains bound to the receptor, returns to surface and is released and used in further cycles^(2,10-11).

The TfR is found in soluble form in human serum and its concentration is related to the contents of receptors expressed in the cells. The soluble TfR (sTfR) is the truncated form of TfR, which lacks the cytoplasmic and transmembrane domains. It is generated by a proteolytic cleavage of TfR, between Arg-100 and Leu-101, by a serine protease located in the endosomal membrane. The receptor circulates as a complex formed by a transferrin molecule and receptor molecule⁽²⁾.

Intracellular iron levels regulate ferritin biosynthesis and TfR expression. This regulation is mediated by the interaction of iron responsive elements (IREs) with cytosol RNA-binding proteins called iron regulatory proteins (IRPs). The IREs are found in untranslated regions (UTRs) of gene mRNAs regulated by iron, such as those of ferritin and transferring receptor. The IRPs are RNA-binding proteins that bind to IREs with high affinity⁽¹²⁾.

The binding of IRPs to IREs leads to changes in expression of iron-regulated genes with subsequent alterations in intracellular iron uptake, use and storage. When iron levels are high, the IREs do not bind to IRPs and the translation of ferritin mRNA occurs normally. Iron deficiency results in IRP binding to IRE in the 5'-UTR of ferritin m-RNA and translation inhibition. Moreover, deficiency leads to IRP binding to IRE in the 3'-UTR of transferrin receptor mRNA

and inhibition of mRNA degradation. The prolonged half-life of mRNA leads to increased TfR expression. The expansion of iron pool inactivates IRP. In the absence of IRP-IRE binding, an endonucleolytic cleavage site is exposed and cleaved, leading to reduced mRNA half-life. Thus, the result of IRPs binding IREs is raised TfR expression in cell surface and decreased ferritin synthesis. These changes lead to increased iron available for use due to greater iron uptake by cells and reduced iron entry for storage. On the other hand, the absence of IRP-IRE binding causes diminished TfR and greater ferritin expression; consequently there is less iron available for cells⁽¹²⁾.

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